

PERISHNET

AI-Powered Predictive Shelf-Life & Post-Harvest Loss Prevention System

Tagline

Scan Once. Monitor Continuously. Prevent Waste.

1. The problem

Fruits and vegetables are **dynamic, perishable inventory**.

Their condition changes continuously because of:

- temperature
- humidity
- time
- storage conditions
- transportation
- handling damage
- ripeness
- initial quality
- produce type

Today, freshness assessment is often:

visual/manual → subjective → periodic → reactive.

A farmer, retailer, warehouse, or household may know:

“This batch looks okay.”

But not:

“How much usable life remains?”

and more importantly:

“What should I do now to prevent this batch from becoming waste?”

That is the problem PerishNet solves.

2. The core idea

PerishNet combines **three sources of intelligence**:

One-time visual inspection

A smartphone photo establishes the initial physical/visual condition.

Continuous environmental sensing

A small sensor pod continuously measures the storage environment.

AI prediction + decision engine

The AI combines the visual condition, environmental history and time to continuously estimate remaining usable life, forecast spoilage risk and recommend the best action.

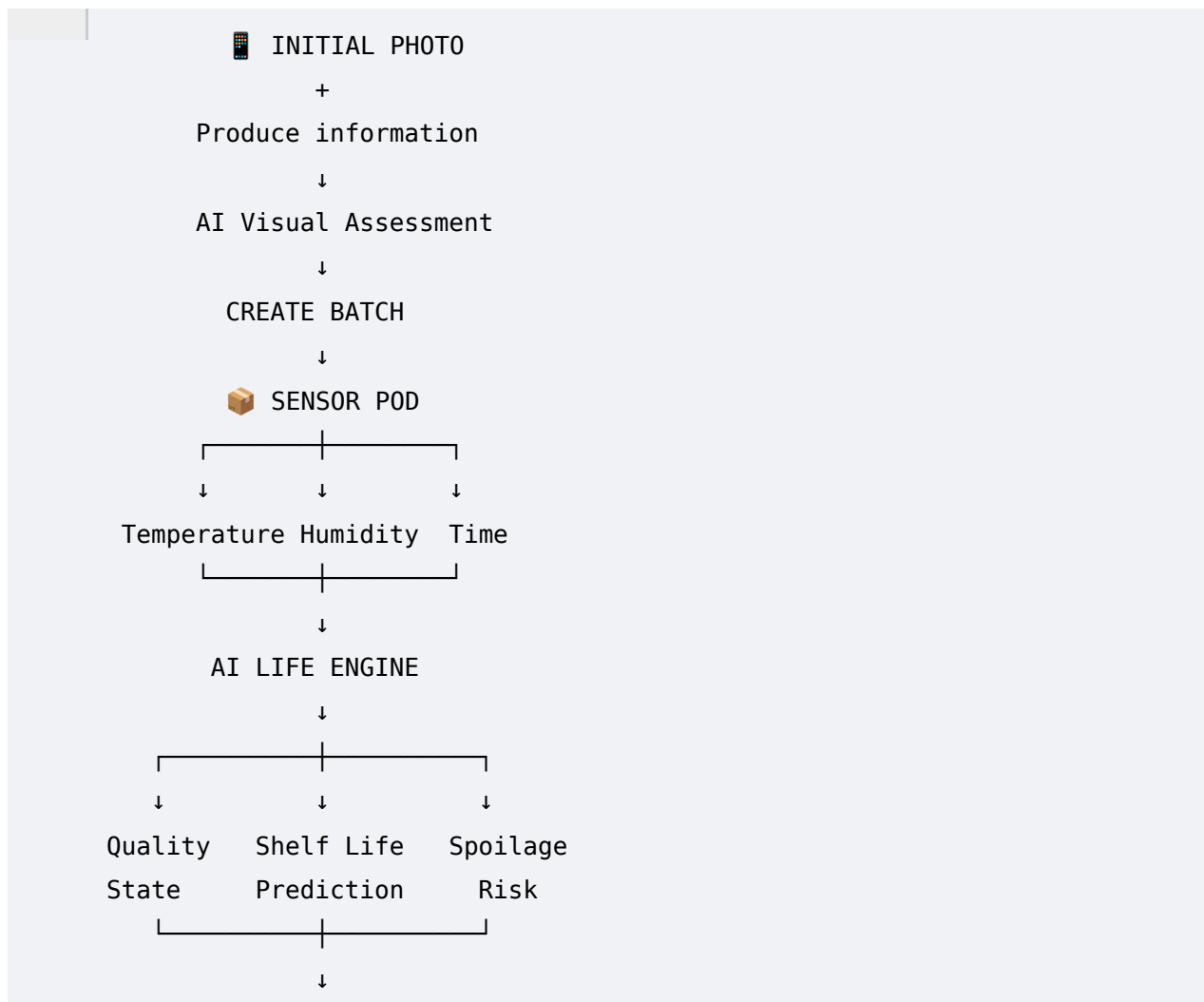
So:

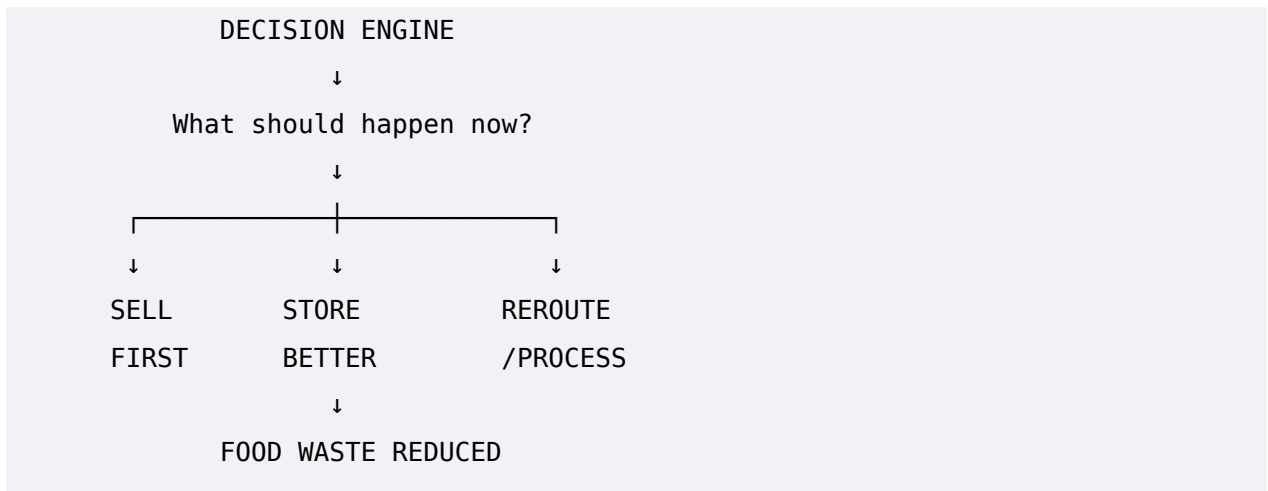
Image tells us where the produce starts.

Sensors tell us what happens afterward.

AI predicts where it will go.

3. The entire flow





4. How the user actually uses it

Step 1 — Create a batch

A user opens the PerishNet app/web app.

They take **one photo** of the produce.

Then enter/confirm:

- produce type
- quantity
- harvest/purchase date
- storage type
- optional location/origin

PerishNet creates:

Batch ID: TOM-042

5. Initial AI inspection

Computer vision analyses:

- ripeness
- visible bruises
- discoloration
- spots
- wrinkles
- surface defects
- mold-like visual patterns

Output:

Produce: Tomato

Condition: Good

Initial quality score: 78/100

Visual confidence: High

But this score is **not the final product**.

It is the starting state for the shelf-life model.

6. Smart Sensor Pod

After creating the batch, the user attaches the small PerishNet device to the appropriate storage environment.

MVP sensors

Temperature

Relative humidity

Time

Optional future sensors

- light exposure
- shock/vibration during transport
- VOC/gas-related sensing
- ethylene-related sensing for suitable produce

Important:

We should **not claim that the device measures every spoilage factor**.

Different produce behaves differently, and gas sensors can be noisy. So the MVP should concentrate on the most practical variables we can measure reliably.

7. Why the sensor is important

Suppose the initial scan says:

Remaining usable life = 4.1 days.

The sensor records:

10:00 → 27.1°C / 67% RH

12:00 → 28.4°C / 70% RH

15:00 → 30.1°C / 74% RH

18:00 → 31.0°C / 77% RH

The AI notices that the produce has experienced increasingly unfavorable conditions.

So its estimate can update:

4.1 days → 3.2 days → 2.6 days

The user doesn't need to photograph the produce every few hours.

8. One scan → continuous intelligence

This is one of your strongest differentiators.

Typical freshness app

Photo → result.

PerishNet

Photo once → establish initial state → continuously monitor environment → dynamically update prediction.

But there's an important safety/accuracy mechanism:

Visual re-verification

Temperature/humidity alone cannot perfectly determine the physical condition of the produce.

So if the model:

- becomes uncertain
- detects an unusual environmental event
- reaches a critical stage

it can request:

 "Please rescan Batch TOM-042."

The new image corrects/recalibrates the prediction.

That gives you a **hybrid AI system**, not a fake "sensor knows everything" system.

9. Dynamic shelf-life prediction ★

This is the main intelligence layer.

Instead of:

“Expires on Tuesday.”

PerishNet predicts:

Estimated remaining usable life: 2.4–3.0 days

based on:

- initial visual condition
- produce type
- batch age
- temperature history
- humidity history
- storage conditions
- other validated features

As conditions change, the forecast changes.

Therefore:

Shelf life is dynamic, not static.

10. Spoilage-risk forecasting

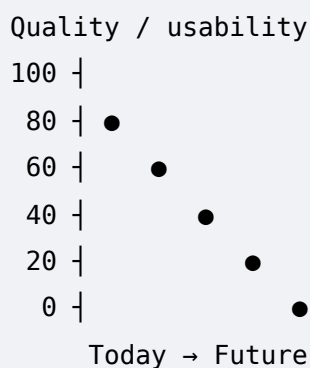
The system doesn't just estimate the remaining life.

It forecasts risk.

Example:

TODAY	● Low
24 HOURS	● Moderate
48 HOURS	● High
72 HOURS	● Critical

The dashboard can show a degradation curve:



This makes the model understandable to judges.

11. Batch-based intelligence ★

This is where PerishNet moves from a consumer app to a real supply-chain solution.

Instead of treating one tomato individually:

Batch TOM-042 — 500 kg

Every batch has:

- initial photo
- quantity
- sensor
- condition history
- predicted shelf life
- risk level
- intervention history
- destination

Now PerishNet can work for:

Farmer

“Should I sell this batch now?”

Warehouse

“Which batch needs cold storage?”

Transporter

“Which batch should travel first?”

Retailer

“Which batch should be sold first?”

Household

“Which vegetables should I consume first?”

12. The decision engine ★★★★★

This is what I would make the **heart of the project**.

Don't stop at:

“This batch has 2 days left.”

Ask:

“What should we do now?”

Example:

Batch	Remaining life
A	6 days
B	2 days
C	4 days

PerishNet:

Prioritize Batch B for sale.

Then it can consider:

- storage capacity
- transport time
- destination
- estimated demand
- remaining usable life

and recommend:

Sell Batch B first.

This is **prediction** → **decision**, not just prediction.

13. Smart routing

Suppose Batch B has:

1.8 days remaining

Market A:

4-hour transport

Market B:

18-hour transport

PerishNet can recommend:

Route Batch B to Market A.

This helps prevent a batch from spending most of its remaining usable life in transportation.

14. Storage recommendation

The AI can also recommend:

Move Batch B to controlled storage.

or:

Current environmental conditions are increasing deterioration risk.

The system can compare:

Current trajectory

vs.

Improved-storage scenario

15. “What-if” simulator ★★★★★

This is one feature I would absolutely include in the SIH demo.

Give the user controls such as:

Temperature

28°C → 34°C

Transport delay

4h → 18h

Storage condition

Current → Controlled

Then show:

Estimated usable life

Spoilage risk

Potential loss

Recommended intervention

The judge can literally manipulate the conditions and watch the prediction change.

This demonstrates that your AI is **modelling the system**, not merely classifying an image.

16. Smart intervention options

PerishNet can recommend among a limited, defensible set of actions:

-  **Continue normal storage**
-  **Prioritize for sale**
-  **Move to better storage**
-  **Reroute for faster distribution**
-  **Send for processing**
-  **Redistribution/recovery pathway**

The exact action set can depend on the produce type, context and safety requirements.

17. QR/NFC digital batch identity

Every batch can have a:

QR code

For example:

 **TOM-042**

Scanning it opens its digital record.

So:

physical produce batch ↔ digital lifecycle

The record contains:

- initial image
- date/time
- sensor history
- environmental graphs
- predicted shelf life
- current risk
- previous rescans
- interventions

- destination

This makes the system traceable.

18. Digital lifecycle timeline

Example:

TOM-042

Day 0

Harvested



Day 1

Initial scan



Day 2

Transport



Day 3



Day 4

PerishNet recommends accelerated sale



Day 5

Batch sold/processed

This is an excellent visual for the PPT.

19. Explainable AI

Never give the user:

“2.37 days.”

with no explanation.

Give:

Why is shelf life falling?

- high ripeness
- increased temperature exposure
- increased humidity
- batch age
- observed visual defects

This makes the system trustworthy and helps during judge Q&A.

20. Confidence and uncertainty

A good scientific system should acknowledge uncertainty.

Instead of:

Remaining life = exactly 2.34 days

show:

Estimated usable life: 2.1–2.8 days

and:

Confidence: Medium

This makes your project more credible.

21. Early-warning system

Use a simple four-stage system:

 **NORMAL**

No immediate intervention.

 **WATCH**

Conditions are becoming unfavorable.

 **ACTION**

Intervention recommended.

 **CRITICAL**

Immediate action / visual verification.

22. Loss-prevention metric

Your primary impact metric should NOT be:

“Our AI has 98% accuracy.”

Accuracy is useful, but the real goal is:

How much loss can we prevent?

Possible dashboard KPIs:

- Expected loss without intervention
- Expected loss with recommended intervention
- Estimated quantity preserved
- Reduction in high-risk batches

Only report these where your model/simulation can substantiate them.

23. Sensor-device architecture

A practical prototype could use:

ESP32

For connectivity and processing.

Temperature/humidity sensor

For environmental monitoring.

Battery

For portable operation.

Bluetooth/Wi-Fi

For communication.

Optional advanced sensors

For specific future produce studies.

The device can send readings to the backend.

Sensor

↓

ESP32

↓

```
BLE / Wi-Fi
↓
PerishNet API
↓
Database
↓
AI Life Engine
```

24. Where the device can be used

Home

Inside/near vegetable storage.

One scan → continuous monitoring.

Retail

Attach to:

- crates
- bins
- storage areas

Warehouse

Many devices:

```
Warehouse
├── Zone A
│   ├── Batch 101
│   └── Batch 102
└── Zone B
    ├── Batch 103
    └── Batch 104
```

The dashboard can identify:

Zone B has abnormal environmental conditions.

Transportation

A future version can monitor conditions during transit.

25. Multi-user platform

Consumer view

Simple:

“Eat tomatoes first.”

Retailer view

Operational:

“Sell Batch B first.”

Warehouse view

Analytical:

“Move these 4 batches to controlled storage.”

Admin view

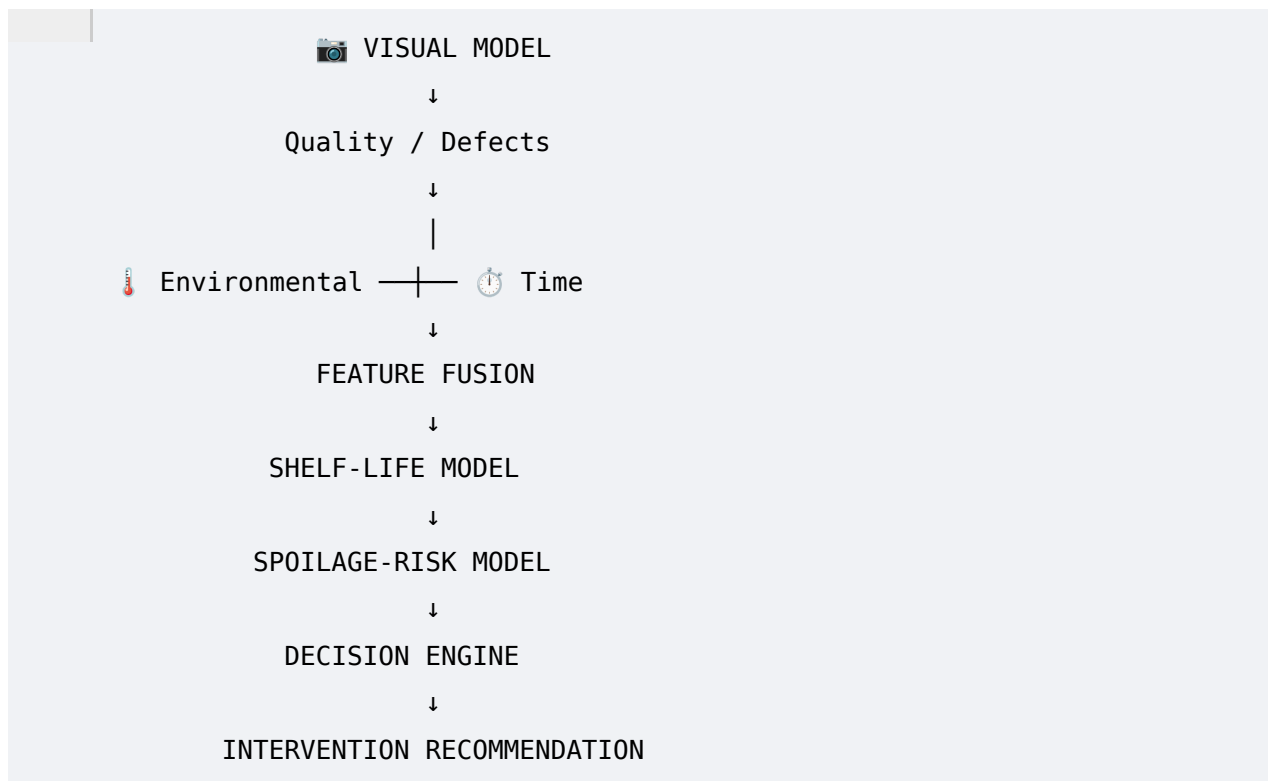
High-level:

“18% of stored inventory currently has elevated loss risk.”

Same underlying system, different interfaces.

26. AI architecture

Your AI can ultimately have multiple components:



Possible models

Computer vision:

- CNN baseline
- EfficientNet
- MobileNet
- Vision Transformer / SegFormer for advanced versions

Shelf-life prediction:

- regression baseline
- gradient boosting
- time-series model
- survival-analysis-style modelling

Decision layer:

- rule-based baseline
- optimization model
- later learned policy if enough data exists

You should **not use every model just to sound advanced**.

Build the simplest model that works, then improve it.

27. Your data strategy

You will need **two types of data**.

Visual dataset

Images showing:

- freshness
- ripeness
- defects
- deterioration stages

Environmental/temporal dataset

Associated:

- temperature
- humidity
- age
- storage condition
- final observed condition

The second one is the harder and more valuable part.

Best approach

Start with public datasets to build the image model.

Then generate your **own experimental dataset** using controlled small-batch observations.

For example:

- 5 produce types
- multiple storage conditions
- repeated measurements
- repeated images
- environmental readings
- final outcomes

That gives you your own research contribution.

28. Very important: don't claim “AI knows expiry”

You should use:

- estimated remaining usable life**

not:

- exact expiry date**

because fruits and vegetables don't have one universally determined expiration time.

Your model is producing a **prediction**, not discovering a hidden printed expiration date.

29. What makes PerishNet different

Don't overload the PPT with 20 “innovations.”

Use these six:

1. One-time visual baseline

A phone creates the initial condition.

2. Continuous environmental sensing

The sensor keeps monitoring the environment.

3. Dynamic shelf-life prediction

Remaining life updates as conditions change.

4. Batch-level intelligence

Designed for real inventory, not just a single fruit.

5. Action recommendation

The system tells the user what to do.

6. What-if simulation

Users can evaluate interventions before acting.

Your core differentiation:

Existing freshness tools primarily answer “How fresh is it?” PerishNet aims to answer “How long will it remain usable, how is that changing, and what should we do now to prevent it from becoming waste?”

That's the sentence I'd memorize.

30. What we should NOT add

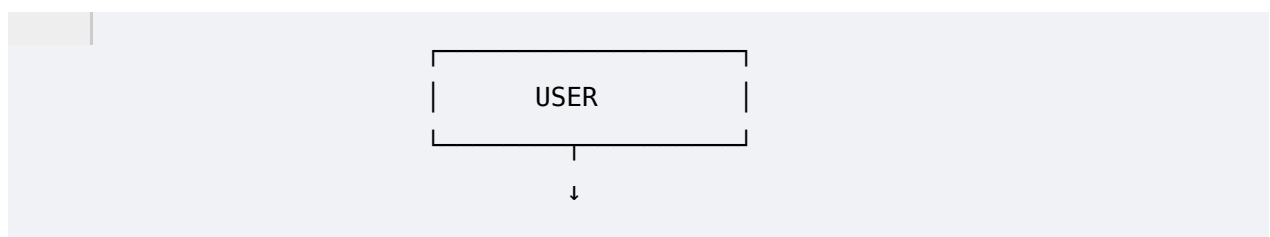
Don't turn the project into:

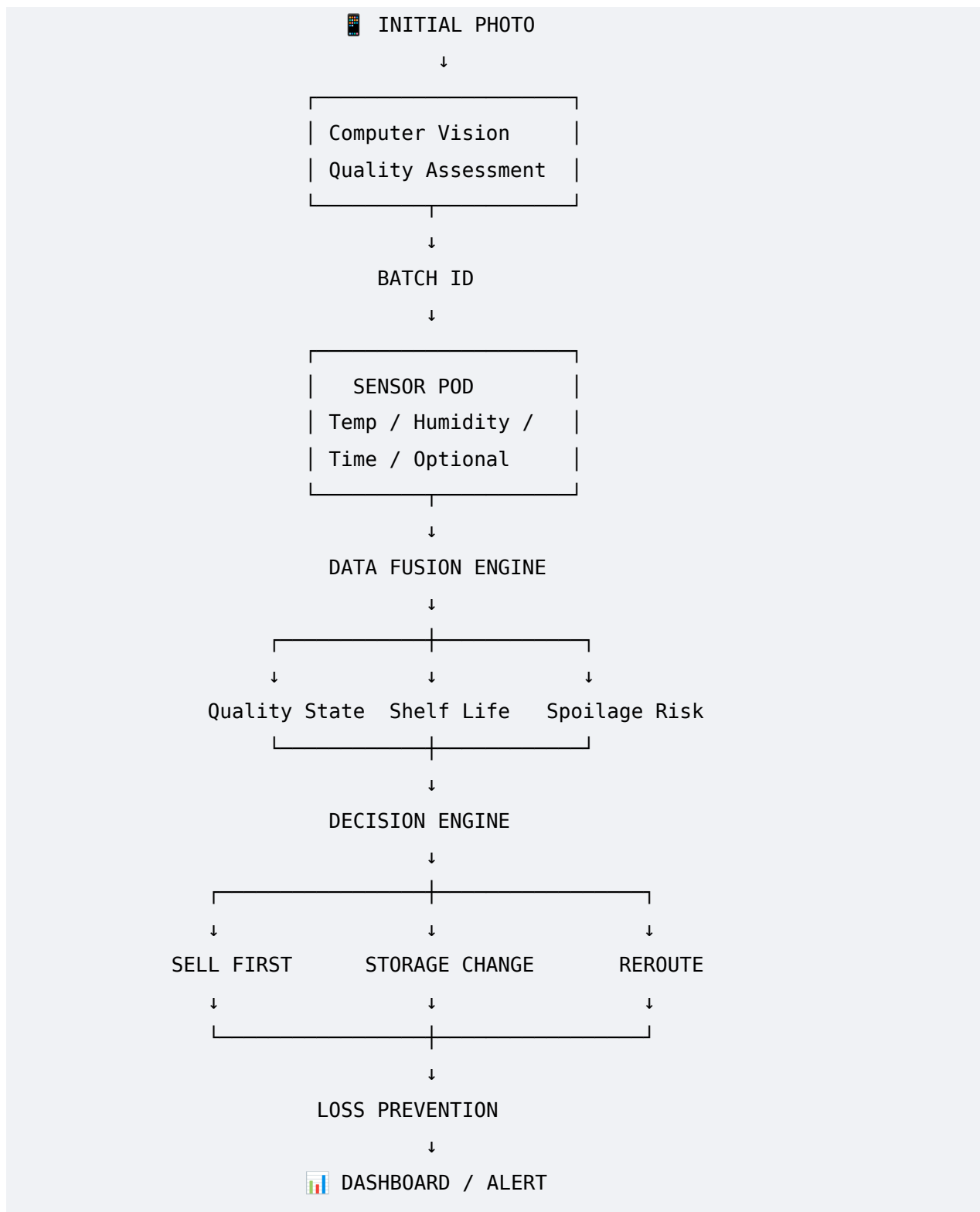
- ✗ blockchain
- ✗ generic chatbot
- ✗ AR/VR
- ✗ marketplace
- ✗ drone
- ✗ unnecessary mobile + website duplication
- ✗ dozens of sensors without validation
- ✗ “AI that detects everything”

Every feature must support:

Assess → Predict → Prevent loss.

31. Final architecture





32. Final product structure

PERISHNET

A. Mobile/Web App

- photo capture
- batch creation

- batch history
- alerts
- recommendations

B. Smart Sensor Pod

- temperature
- humidity
- time
- connectivity

C. AI Engine

- visual assessment
- shelf-life prediction
- risk forecasting
- intervention recommendation

D. Supply-Chain Dashboard

- batch monitoring
- warehouse zones
- inventory priority
- alerts
- what-if simulation

33. The SIH-level problem statement for your PPT

Perishable fruits and vegetables continuously lose quality after harvest, but conventional inventory and storage systems generally lack a dynamic understanding of their remaining usable life. This can lead to premature spoilage, inefficient storage, poor prioritization, and avoidable post-harvest losses.

PerishNet proposes an AI-powered predictive system that combines an initial smartphone-based visual assessment with continuous environmental sensing to dynamically estimate remaining usable life, forecast spoilage risk, and recommend timely interventions such as prioritized sale, improved storage, rerouting, processing or redistribution.

34. One-line solution

PerishNet converts perishable produce from static inventory into continuously monitored, predictive digital batches.

That is a very strong line for the PPT.

35. Your 10-slide PPT structure

Slide 1 — Title

PERISHNET

Scan Once. Monitor Continuously. Prevent Waste.

Slide 2 — Problem

Why post-harvest perishability is difficult to manage.

Slide 3 — Why current approaches fall short

Manual inspection / static expiry / basic freshness classifiers.

Slide 4 — Our solution

One photo + sensor pod + AI life engine.

Slide 5 — How PerishNet works

The complete pipeline.

Slide 6 — Core AI

Visual assessment + sensor fusion + dynamic shelf-life prediction.

Slide 7 — Decision intelligence

Sell / store / reroute / process + what-if simulator.

Slide 8 — Smart hardware

Sensor pod + QR/NFC batch identity.

Slide 9 — Demo / impact

Show one batch going:

Healthy → risk → intervention → loss prevented

Slide 10 — Scalability & future

Home → retailer → warehouse → transportation → supply chain.

36. The strongest demo scenario

This should be your centerpiece.

Initial scan

 **Tomato Batch T-042**

Quality: Good

Remaining life: 4.2 days

Then the sensor detects increasing temperature.

The forecast changes:

Remaining life: 2.6 days

PerishNet says:

 **Action recommended: prioritize this batch for sale.**

You then simulate:

“What if we move it to controlled storage?”

The forecast improves.

Then:

“What if transportation is delayed 18 hours?”

Risk increases.

The judge immediately sees:

monitoring → prediction → simulation → intervention.

37. Your team's core philosophy

Do not think:

“We are making an AI app for fruits.”

Think:

“We are building an intelligence layer for perishable inventory.”

That mindset will guide every feature and prevent the project from becoming a collection of disconnected ideas.

38. Final version — one sentence

PerishNet is an AI-powered smart sensing and decision-support platform that uses one-time visual inspection and continuous environmental monitoring to

dynamically predict the remaining usable life of perishable produce and recommend interventions that reduce avoidable food loss